INCH-POUND

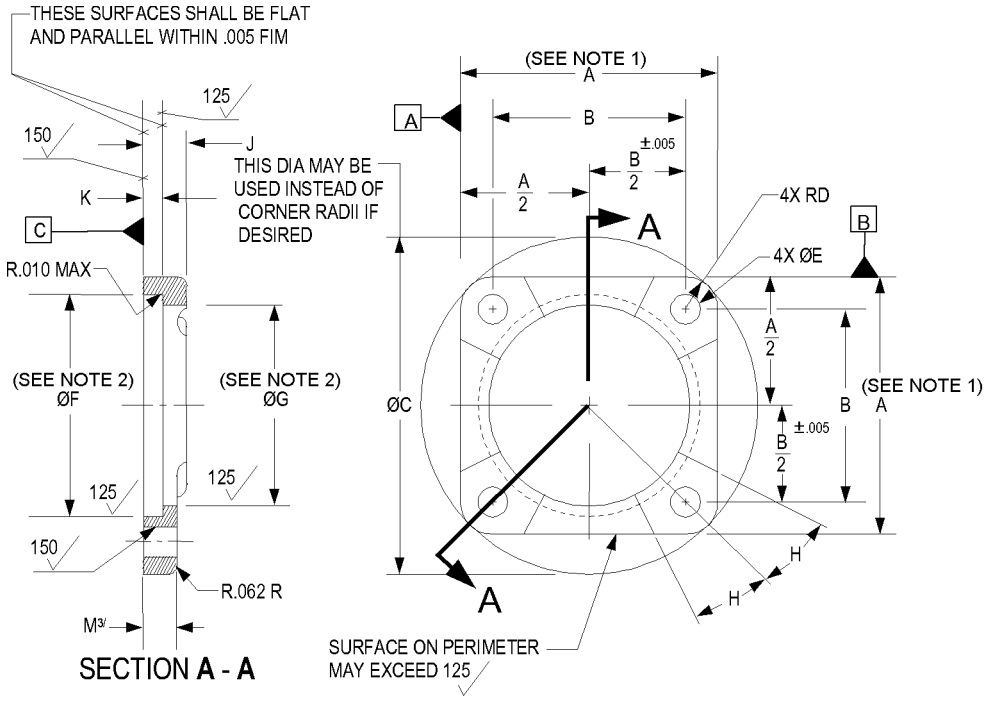
MS27077C
25 September 2003
SUPERSEDING
MS27077B
2 May 1966

DETAIL SPECIFICATION SHEET

FLANGE, SWIVEL, RETAINING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-27272.



NOTES:
1. Reduction of dimensions by 7º max draft angle permissible if forged.
2. Diameters F and G shall be concentric within .010 full indicator movement (FIM).

FIGURE 1. Flange illustration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:15Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Intended use. This part is a component of MS27062, MS27064, and MS27066. This is a design standard for manufacturing purposes. The item is only procured as an integral part of adapter assemblies.

Identification of product. The Part or Identifying Number (PIN) for this part shall be as specified in table I (e.g., MS27077-8C).

Dimensions and tolerances. Dimensions are in inches. Unless otherwise specified, break or radius all corners .005, +.005, -.000.

Material. Corrosion-resistant steel, class 304, condition A, in accordance with SAE AMS-QQ-S-763.

Finish. Passivate in accordance with SAE AMS-QQ-P-35.

Remove all burrs and slivers.

Flange illustration. See figure 1.

Surface roughness. Unless otherwise specified, maximum surface roughness shall not exceed 125 µin. R_a in accordance with ASME B46.1.

Order of precedence. This specification takes precedence over the documents referenced herein. Unless otherwise specified, referenced documents shall be of the issue in effect on the date of solicitation.

TABLE I. Flange requirements.

PIN MS27077	A 1/		B ±.005	C ±.015	D ±.016	E +.010 -.000
-8C	1.390	±.016	.950	1.782	.219	.205
-10C	1.468		1.038	1.906		
-12C	1.594		1.156	2.094		
-16C	1.750		1.312	2.312		
-20C	2.188	±.020	1.656	2.875	.266	.266
-24C	2.375		1.812	3.094	.281	

PIN MS27077	F 2/ +.005 -.000	G 2/ +.010 -.000	H ±.5º	J .016	K +.005 -.000	M 3/ +.015 -.000
-8C	.885	.750	20º	.297	.136	.234
-10C	1.010	.880				
-12C	1.260	1.125				
-16C	1.510	1.375	18º			
-20C	1.854	1.688	16º			
-24C	2.135	1.938		.359	.168	.297

1/ Reduction of dimensions by 7º max draft angle permissible if forged.

2/ Diameters F and G shall be concentric within .010 full indicator movement (FIM).

3/ Part may be forged or machined from bar. If machined, thickness shall be to dimension M.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:15Z
Check the source to verify that this is the current version before use.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army –AV
Navy - AS
Air Force - 99
DLA - CC

Review activities:
Army - AR, AT, MI
Navy - MC, SA, SH
Air Force - 71

Preparing activity:
DLA - CC

(Project 4730-0868-080)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:15Z
Check the source to verify that this is the current version before use.